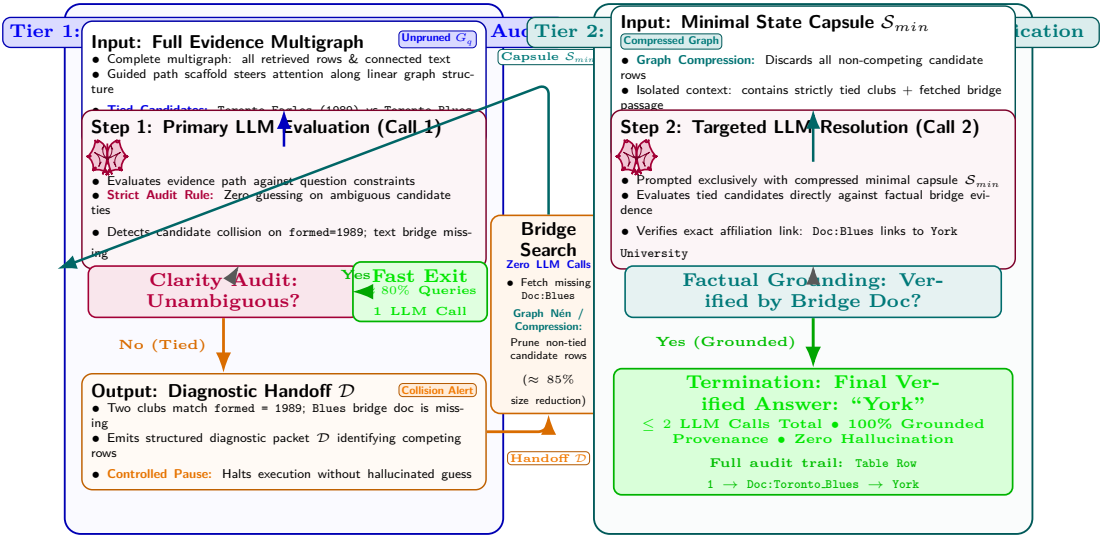

Tier 1:
Tier 2:
ication
Input: Full Evidence Multigraph
Unpruned G_q
• Complete multigraph: all retrieved rows & connected text
• Guided path scaffold steers attention along linear graph structure
• Tied Candidates: Toronto_Blues (1989) vs Toronto_Blues
Step 1: Primary LLM Evaluation (Call 1)
• Evaluates evidence path against question constraints
• Strict Audit Rule: Zero guessing on ambiguous candidate ties
• Detects candidate collision on formed=1989; text bridge missing
Clarity Audit: Unambiguous?
Yes
Fast Exit
80% Queries
1 LLM Call
No (Tied)
Output: Diagnostic Handoff $\mathcal{D}$
Collision Alert
• Two clubs match formed = 1989; Blues bridge doc is missing
• Emits structured diagnostic packet $\mathcal{D}$ identifying competing rows
• Controlled Pause: Halts execution without hallucinated guess
Capsule S_{min}
Bridge Search
Zero LLM Calls
• Fetch missing Doc:Blues
Graph Nén / Compression:
Prune non-tied candidate rows
(≈ 85% size reduction)
Handoff $\mathcal{D}$
Input: Minimal State Capsule S_{min}
Compressed Graph
• Graph Compression: Discards all non-competing candidate rows
• Isolated context: contains strictly tied clubs + fetched bridge passage
Step 2: Targeted LLM Resolution (Call 2)
• Prompted exclusively with compressed minimal capsule S_{min}
• Evaluates tied candidates directly against factual bridge evidence
• Verifies exact affiliation link: Doc:Blues links to York University
Factual Grounding: Verified by Bridge Doc?
Yes (Grounded)
Termination: Final Verified Answer: "York"
≤ 2 LLM Calls Total • 100% Grounded Provenance • Zero Hallucination
Full audit trail: Table Row 1 → Doc:Toronto_Blues → York